CSA4704-DEEP LEARNING

NAME:HARITHI S **REG NO:192421151**

Experiment 23: Bayesian Network Construction and Visualization

Problem Statement: A clinic wants a decision-support tool that reasons about the causes of a patient's symptoms under uncertainty. A medical diagnosis network with nodes Smoking, Pollution, Lung Disease, Cough, X-ray Result and Fatigue, with given conditional probability tables. The objective is to build a Bayesian network, define its CPDs and visualise its structure. Use pgmpy, NetworkX, Matplotlib.

Tasks:

1. Define the directed acyclic graph in pgmpy.

2. Specify the conditional probability distribution for every node.

3. Validate the model with check_model().

4. Draw the network with NetworkX.

5. List the conditional independencies and active trails in the network.

FORMULA:

1. Joint Probability Distribution

$P(S, P, L, C, X, F) = P(S) \times P(P) \times P(L|S,P) \times P(C|L) \times P(X|L) \times P(F|L)$

2. Conditional Probability

$P(A|B) = P(A \cap B) / P(B)$

3. Bayes' Theorem

$P(A|B) = [P(B|A) \times P(A)] / P(B)$

4. Marginal Probability

$P(A) =$ Sum of $P(A,B)$ over all values of B

5. Conditional Probability Distribution (CPD)

$P(X|Parents(X))$

6. Independence

$P(A,B) = P(A) \times P(B)$

7. Conditional Independence

$P(A|B,C) = P(A|C)$

8. Probability Normalization

$P(True) + P(False) = 1$

9. Posterior Probability

$P(Disease|Symptoms) = P(Disease, Symptoms) / P(Symptoms)$

10. Joint Probability

$P(A,B,C) = P(A) \times P(B|A) \times P(C|B)$

CODE:

```
!pip install -q pgmpy networkx matplotlib

import networkx as nx

import matplotlib.pyplot as plt

from pgmpy.models import DiscreteBayesianNetwork
```

```
from pgmpy.factors.discrete import TabularCPD

# 1. Define the directed acyclic graph (DAG)

model = DiscreteBayesianNetwork([

    ("Smoking", "LungDisease"),

    ("Pollution", "LungDisease"),

    ("LungDisease", "Cough"),

    ("LungDisease", "XrayResult"),

    ("LungDisease", "Fatigue")

])

# 2. Define conditional probability distributions (CPDs)

# State 0 = No/Low/Negative; State 1 = Yes/High/Positive


cpd_smoking = TabularCPD(

    variable="Smoking",

    variable_card=2,

    values=[[0.7], [0.3]]

)

cpd_pollution = TabularCPD(

    variable="Pollution",

    variable_card=2,
```

```
    values=[[0.8], [0.2]]
)
# Columns: Smoking=0,Pollution=0;
# Smoking=0,Pollution=1;
# Smoking=1,Pollution=0;
# Smoking=1,Pollution=1
cpd_lung = TabularCPD(
    variable="LungDisease",
    variable_card=2,
    values=[
        [0.99, 0.96, 0.95, 0.85],
        [0.01, 0.04, 0.05, 0.15]
    ],
    evidence=["Smoking", "Pollution"],
    evidence_card=[2, 2]
)
cpd_cough = TabularCPD(
    variable="Cough",
    variable_card=2,
    values=[
```

```
        [0.9, 0.2],
        [0.1, 0.8]
    ],
    evidence=["LungDisease"],
    evidence_card=[2]
)
cpd_xray = TabularCPD(
    variable="XrayResult",
    variable_card=2,
    values=[
        [0.95, 0.1],
        [0.05, 0.9]
    ],
    evidence=["LungDisease"],
    evidence_card=[2]
)

cpd_fatigue = TabularCPD(
    variable="Fatigue",
    variable_card=2,
```

```
    values=[
        [0.8, 0.3],
        [0.2, 0.7]
    ],
    evidence=["LungDisease"],
    evidence_card=[2]
)

model.add_cpds(
    cpd_smoking,
    cpd_pollution,
    cpd_lung,
    cpd_cough,
    cpd_xray,
    cpd_fatigue
)

# 3. Validate the Bayesian network
print("Is the model valid?", model.check_model())
print("\nConditional Probability Distributions:")
```

```
for cpd in model.get_cpds():
    print("\n", cpd)
# 4. Draw the network using NetworkX
G = nx.DiGraph()
G.add_edges_from(model.edges())


pos = {
    "Smoking": (0, 2),
    "Pollution": (2, 2),
    "LungDisease": (1, 1),
    "Cough": (0, 0),
    "XrayResult": (1, 0),
    "Fatigue": (2, 0)
}
plt.figure(figsize=(10, 6))
nx.draw_networkx(
    G,
    pos=pos,
    with_labels=True,
    node_size=3000,
```

```
    node_color="lightblue",
    font_size=9,
    font_weight="bold",
    arrows=True,
    arrowsize=20
)
plt.title("Medical Diagnosis Bayesian Network")
plt.axis("off")
plt.show()
# 5. Conditional independencies
print("\nIMPORTANT CONDITIONAL INDEPENDENCIES:")
print("Smoking and Pollution are independent root nodes.")
print("Cough, XrayResult and Fatigue are conditionally")
print("independent of each other given LungDisease.")
# Ask pgmpy to display its implied independencies
print("\nIndependencies implied by the graph:")
try:
    print(model.get_independencies())
except (AttributeError, TypeError) as error:
    print("Use the important independencies listed above.")
```

```
    print("Details:", error)


# 6. Active trails
print("\nACTIVE TRAILS FROM Smoking:")
try:
    print("Without observations:")
    print(model.active_trail_nodes("Smoking", observed=[]))
    print("\nWhen LungDisease is observed:")
    print(model.active_trail_nodes(
        "Smoking", observed=["LungDisease"]
    ))
except (AttributeError, TypeError) as error:
    print("Active-trail API differs in this pgmpy version.")
    print("Details:", error)
print("\nExperiment 23 completed successfully.")
```

OUTPUT:

```
Is the model valid? True


Conditional Probability Distributions:
```

Smoking(0)	0.7
Smoking(1)	0.3

Pollution(0)	0.8
Pollution(1)	0.2

Smoking	Smoking(0)	Smoking(0)	Smoking(1)	Smoking(1)
Pollution	Pollution(0)	Pollution(1)	Pollution(0)	Pollution(1)
LungDisease(0)	0.99	0.96	0.95	0.85
LungDisease(1)	0.01	0.04	0.05	0.15

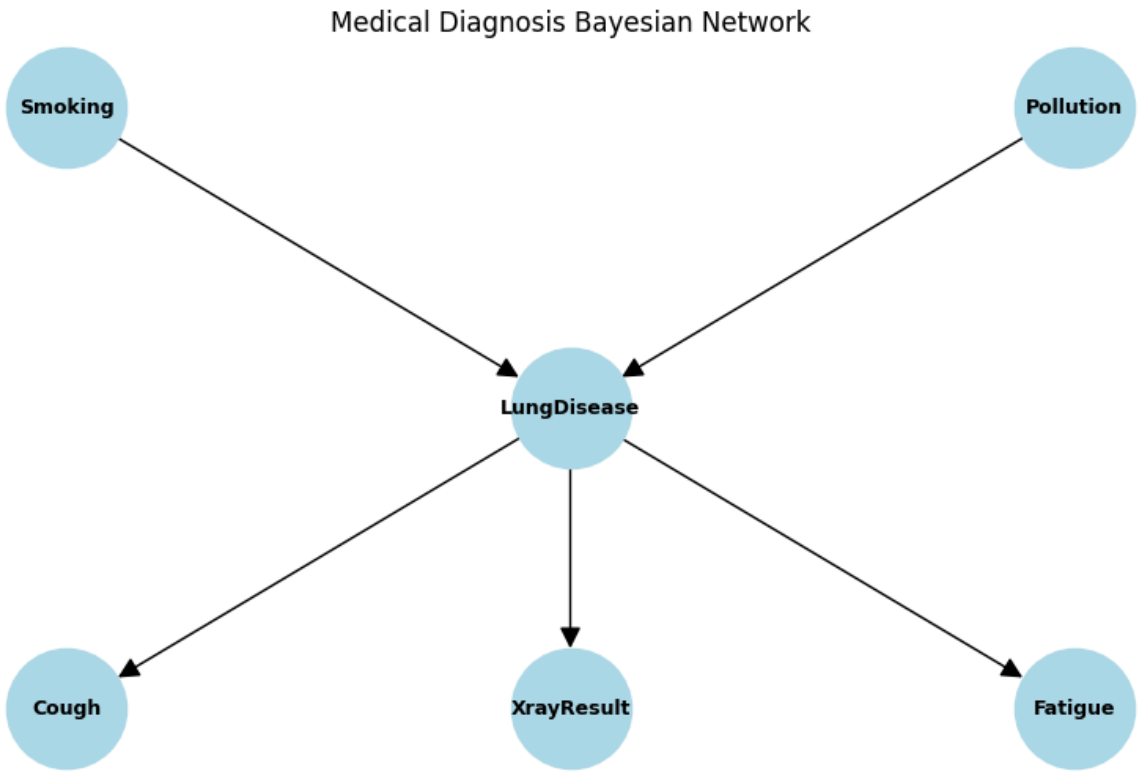
LungDisease	LungDisease(0)	LungDisease(1)
Cough(0)	0.9	0.2
Cough(1)	0.1	0.8

LungDisease	LungDisease(0)	LungDisease(1)
XrayResult(0)	0.95	0.1
XrayResult(1)	0.05	0.9

LungDisease	LungDisease(0)	LungDisease(1)
Fatigue(0)	0.8	0.3

| Fatigue(1) | 0.2 | 0.7 |

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IMPORTANT CONDITIONAL INDEPENDENCIES:

Smoking and Pollution are independent root nodes.

Cough, XrayResult and Fatigue are conditionally

independent of each other given LungDisease.

Independencies implied by the graph:

(Cough ⊥ XrayResult | LungDisease)

(Fatigue ⊥ Pollution | LungDisease)

(Fatigue ⊥ Smoking | LungDisease)

(Pollution ⊥ XrayResult | LungDisease)

(Cough ⊥ Smoking | LungDisease)

(Fatigue ⊥ XrayResult | LungDisease)

(Cough ⊥ Fatigue | LungDisease)

(Pollution ⊥ Smoking)

(Smoking ⊥ XrayResult | LungDisease)

(Cough ⊥ Pollution | LungDisease)

ACTIVE TRAILS FROM Smoking:

Without observations:

{'Smoking': {'Cough', 'Smoking', 'LungDisease', 'Fatigue', 'XrayResult'}}

When LungDisease is observed:

{'Smoking': {'Smoking', 'Pollution'}}